

- Smart voting in reality shows
- Electronic health records
- Real estate and assets management
- Logistics and supply chain management

According to *Statista Research*, the market for blockchain related products and services has been expanding rapidly over the past several years, and will be worth close to US\$ 40 billion by 2025.

Open platforms for smart contracts programming

The development of blockchain apps is focused on decentralized applications with the implementation of smart contracts programming. A decentralized application (dApp) refers to the specialised software that runs across several nodes connected by a network, making it extremely difficult to hack. The code in a conventional centralised setup resides on a single server. The biggest problem with centralised systems is that everything is at risk of being lost or duplicated if the central server is hacked. In a decentralized application, data is stored locally on each user's device rather than on a central server. This ensures that there will always be a copy of each transaction for replication purposes. Hacking a decentralized

Frameworks, toolkits and prominent libraries for a blockchain environment

Platform/Toolkit	URL
Truffle	truffleframework.com/
Dot-Abi-cli	github.com/cryppadotta/dotta-license/tree/master/dot-abi-cli
Remix IDE	remix.ethereum.org/
Embark	embark.status.im/
Populus	populus.readthedocs.io/en/latest/
PyEthereum	github.com/ethereum/pyethereum
Etherlime	etherlime.readthedocs.io/en/latest/
EthFiddle	ethfiddle.com/
Netherium	netherium.com/
Go Ethereum / Geth	geth.ethereum.org/ github.com/ethereum/go-ethereum/wiki/geth
Openchain	openchain.org/
Cava	github.com/consensys/cava
Web3JS	github.com/ethereum/web3.js
Solidity	solidity.readthedocs.io/en/v0.4.24/
CPP Ethereum	ethdocs.org/en/latest/ethereum-clients/cpp-ethereum/
Liquidity	liquidity-lang.org/
Infura	infura.io/
Lamden	lamden.io/
MyThril	consensys.net/diligence/mythril.html
Coq	coq.inria.fr/
Hydrachain	github.com/HydraChain/hydrachain
Multichain	multichain.com/
Ethereum	ethereum.org/
Corda	corda.net/
Credits	credits.com/
BigchainDB	bigchaindb.com/
Quorum	jpmorgan.com/global/Quorum
Symbiont Assembly	symbiont.io/
Embark	embark.status.im/

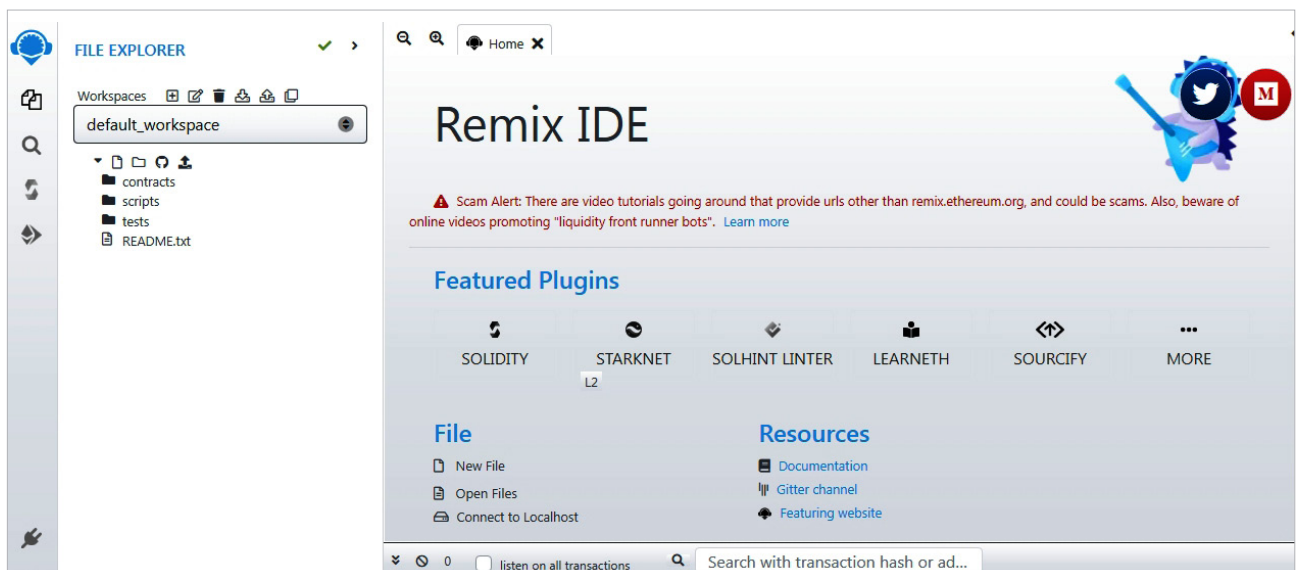


Figure 1: Smart contracts development platform of Remix IDE